Very Low-Resolution Moving Vehicle Detection in Satellite Videos

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***Abstract*— This article proposes a practical end-to-end neural network framework to detect tiny moving vehicles in satellite videos with low imaging quality. Some instability factors, such as illumination changes, motion blurs, and low contrast to the cluttered background, make it difficult to distinguish true objects from noise and other point-shaped distractors. Moving vehicle detection in satellite videos can be carried out based on background subtraction or frame differencing. However, these methods are prone to produce lots of false alarms and miss many positive targets. Appearance-based detection can be an alternative but is not well-suited since classifier models are of weak discriminative power for the vehicles in top view at such low resolution. This article addresses these issues by integrating motion information from adjacent frames to facilitate the extraction of semantic features and incorporating the transformer to refine the features for key points estimation and scale prediction. Our proposed model can well identify the actual moving targets and suppress interference from stationary targets or background. The experiments and evaluations using satellite videos show that the proposed approach can accurately locate the targets under weak feature attributes and improve the detection performance in complex scenarios.**

Index Terms— **Detection, end-to-end neural network framework, integrated motion information, low-resolution, moving vehicle, satellite video, transformer.**

Manuscript received March 7, 2022; revised May 14, 2022; accepted May 28, 2022. Date of publication June 1, 2022; date of current version June 13, 2022. This work was supported in part by the Key Scientific Technological Innovation Research Project by Ministry of Education; in part by the State Key Program and the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61836009 and Grant 61621005; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-06; in part by the Major Research Plan of the National Natural Science Foundation of China under Grant 91438201, Grant 91438103, and Grant 61801124; in part by the National Natural Science Foundation of China under Grant U1701267, Grant 62006177, Grant 61871310, Grant 61902298, Grant 61573267, Grant 91838303, and Grant 61906150; in part by the Fund for Foreign Scholars in University Research and Teaching Program's 111 Project under Grant B07048; in part by the Program for Cheung Kong Scholars and Innovative Research Team in University under Grant IRT 15R53; in part by the ST Innovation Project from the Chinese Ministry of Education; in part by the National Science Basic Research Plan in Shaanxi Province of China under Grant 2019JQ-659; in part by the 2018 Postdoctoral Foundation of Shaanxi Province under Grant 2018BSHEDZZ46; in part by the Scientific Research Project of Education Department in Shaanxi Province of China under Grant 20JY023 and Grant 2022JQ-607; in part by the Fundamental Research Funds for the Central Universities under Grant XJS201901, Grant XJS201903, Grant JBF201905, and Grant JB211908; and in part by the CAAI-Huawei MindSpore Open Fund. *(Corresponding author: Licheng Jiao.)*

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Digital Object Identifier 10.1109/TGRS.2022.3179502

I. Introduction

SATELLITE video can acquire a period of continuous observation over a certain area, which extracts both static and dynamic information on the ground. It has great potential for numerous applications in both the civilian and military domains. Compared to the ground-fixed or aerial-based camera video, it can capture objects at large-scale scenes, which has become a new powerful way for city-scale traffic surveillance, driver behavior analysis, traffic density monitoring, or urban management [1]–[3]. It may become the main data source for future intelligent transportation systems [4]. In addition, this real-time satellite imagery enables a lot of potential applications in military areas, including military reconnaissance, detection of abnormal or dangerous behavior, and surveillance of restricted areas [5]. As the foundation of intelligent traffic monitoring or military surveillance, moving vehicle detection in satellite videos has been well-studied, which is unavailable in traditional remote sensing satellite images.

The SkyBox Imaging (USA) launched SkySat-1 (in 2013), which could capture high-definition (HD) grayscale videos with ground sample distances (GSDs) of around 1 m [3], [6], [7]. In 2015, Chang Guang Satellite Technology Company Ltd (CGSTL) launched two video satellites Jilin1-1 and Jilin1-2, which could shoot full-color videos with a resolution of nearly 1 m [3], [8]. Based on such satellite videos, due to the GSDs being no greater than 1 m, moving vehicles are captured with only several or tens of pixels spatially. Even if some conventional methods perform well in ground-based camera videos, they probably fail to identify on satellite videos. The reason may be attributed to the following three unique characteristics.

1) *Limited Spatial Features:* Since the lack of spatial features, such as shapes and textures, the motion information of vehicles may be the most or only useful feature to some extent.
2) *Low Contrast to the Background:* As shown in Fig. 1, most moving vehicles appear as white spots, which are hard to be identified from the low-contrast roads.
3) *Nonstationary Platform:* The satellite videos exhibit a unique type of local misalignment and intensity

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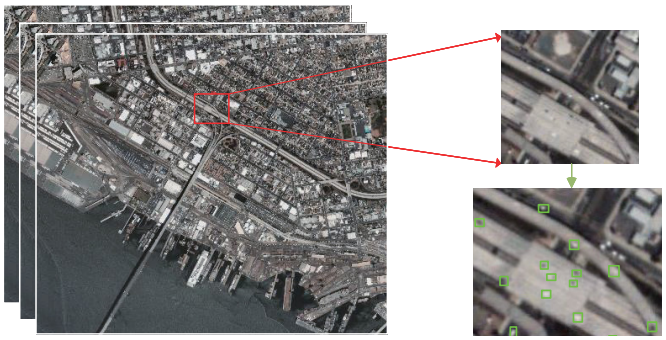


Fig. 1. Examples to illustrate the moving vehicles in a local scene of video captured by the JiLin satellite. (Top right) Enlarged region of the red local scene. (Bottom right) Corresponding ground truth annotations of moving vehicles.

variations [8]. The local variation of pixels results in one of the primary sources of false alarms.

Traditional research on object detection for satellite video currently relies on background subtraction and frame differencing [3], [8]–[15]. However, these motion-based approaches are prone to inadequate background modeling and suffer heavily from false positives (FPs) introduced by the problem of parallax effects, local misalignment [8], and illumination variations. Recent literature [16]–[21] also suggests applying prevailing deep learning (DL) on aerial or satellite still images, which could accomplish the detection of the moving targets frame by frame. Nevertheless, the works show moderate performance for minimal feature information. In addition, these exiting DL methods need to be combined with background subtraction models (BGM) [6], [17] or some postprocessing steps [5], [22]. Detecting such small vehicles from satellite videos directly by DL detectors is challenging.

Regarding the problems mentioned above, the performance involves two main issues: object localization and false target removal. Therefore, to improve the detector's discrimination ability and suppress interference from the cluttered background, we propose a practical end-to-end method to predict the location of moving objects from satellite videos. Due to the lack of general appearance features, the motion information from adjacent frames is provided to assist detection and facilitate the extraction of semantic features. In addition, considering motion blurs and low contrast to the background, the concept of information compensation and noise suppression should be employed. To this end, we establish a differential module in the feature extraction stage to weaken the features of the background area and enhance the moving target based on the operation of correlation.

Although the background area has been suppressed to a certain extent by the differential module, the captured satellite videos exhibit a unique type of local misalignment and intensity variations—small and uneven motions on some stationary background objects [8], which results in one of the main challenge in moving object detection. Therefore, transformer [23], [24] is introduced to refine the feature from the differential module, which could capture short- and long-range visual dependencies. Based on the self-attention mechanism, we perform pixel-level classification and regression on the refined feature map. Considering the resource consumption of transformer and the sparsity of targets in remote sensing images, we propose a unique attention layer to replace the traditional multihead attention (MHA) layer [23] in the encoder. The overall flowchart of the proposed moving vehicles detection framework is shown in Fig. 2. The main contributions of this article are summarized as follows.

1) Due to the lack of useful appearance features, we design a differential module in the stage of feature extraction. It integrates the motion information from adjacent frames to facilitate the extraction of effective semantic features. In addition, considering the interference from static objects and the cluttered background, the module can suppress the response of the unrelated areas by differencing operations.
2) Transformer is introduced to refine the semantic features based on the self-attention mechanism. To alleviate the high computation consumption, we propose a unique attention layer to replace the traditional MHA. Since the sparsity of targets in remote sensing images, interactions would be concentrated on the responses of truly moving vehicles.
3) We developed a practical framework that can be trained in an end-to-end manner without any postprocessing. To evaluate the proposed approach, we conducted extensive experiments on a collection of satellite videos and compared it with state-of-the-art algorithms. The results demonstrate the superior performance of our model over the others on moving vehicles detection.

The remainder of this article is organized as follows. Related works are presented in Section II. We elaborate on the proposed approach for moving vehicles detection in Section III, which is followed by the analysis of the parameter settings and the experimental results in Section IV. Finally, conclusions and suggestions for future research are offered in Section V.

II. Related Work

Nowadays, many remote sensing observation technologies have been developed, and a lot of static or dynamic images can be obtained from spaceborne or airborne camera platforms, including optical satellites, spaceborne synthetic aperture radar (SAR) [25], [26], and aircraft [3]. Detection algorithm based on these data has achieved considerable success for its broad application in the city-scale traffic surveillance system and urban management.

BGM is a commonly adopted idea for detecting moving objects [2], [8]–[10], [27], [28], which aims to recognize the moving foreground from the difference between the current frame and an acquired background (reference) frame of the video. Zhang *et al.* [8] proposed a novel BGM with moving-confidence-assisted low-rank matrix decomposition and foreground regularization for moving vehicles detection of satellite video. Ahmadi and Mohammadzadeh [29] employed background subtraction and per frame Otsu thresholding to detect vehicles and vessels from high-resolution spaceborne videos. Friedman and Russell [30] proposed the Gaussian mixture

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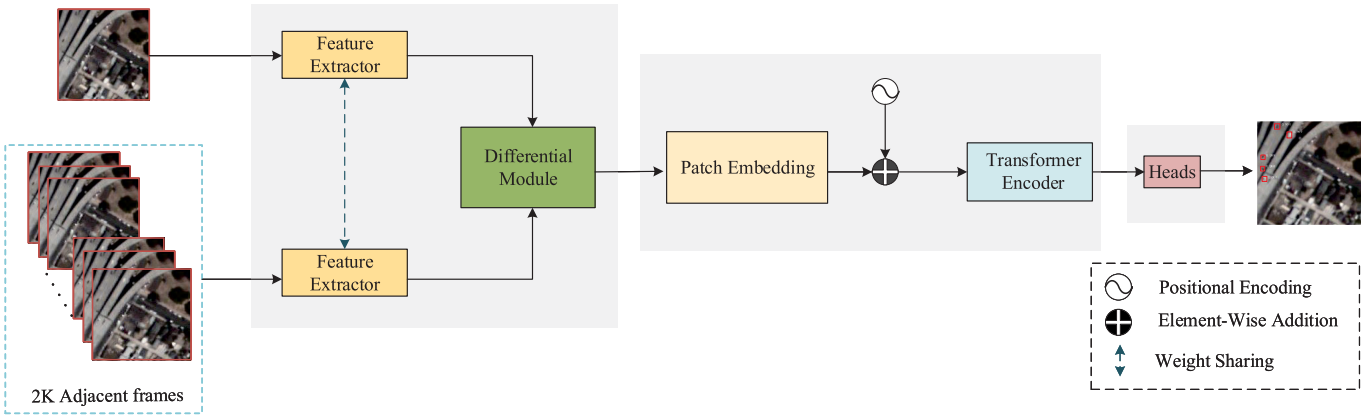


Fig. 2. Proposed framework. It contains three main components: 1) the differential module to weaken the interference of background and enhance the semantic feature; 2) the transformer encoder to refine the feature; and 3) the head of classifier and bounding box regression.

model (GMM), which modeled each background pixel by mixing K (from 3 to 5) Gaussian distributions. Subsequently, studies [10], [31], [32] have been done to improve and extend the GMM model. However, BGM depends heavily on illumination changes and parallax effect, which are very common in satellite videos for the significant variations of the received signals [8] and the top view shooting attitude.

Frame difference method is an effective moving object detection algorithm with low computational cost, which identifies moving objects by calculating the difference between adjacent frames. Xiao *et al.* [12] accomplished vehicle detection and tracking based on frame difference in wide-area aerial video. Saur *et al.* [15] introduced an effective image differencing method to suppress unrelated motion by using the minimum differences of each pixel in small neighborhoods. Keck *et al.* [33] employed a box filter to locally average the difference image and this filtered value was subtracted from the initial difference image, which could reduce FP detections in the aerial imagery to some extent. Du *et al.* [34] proposed a new object location method by fusing the kernel correlation filter (KCF) tracker and the three-frame-difference method for satellite video. Nevertheless, these algorithms are also sensitive to noise and illumination changes, resulting in many false alarms.

DL has exhibited a powerful ability in many studies [5], [18]–[22]. Some DL-based algorithms have also been proposed as dominant frameworks to deal with moving object detection and tracking. Guo *et al.* [19] proposed a unified multiscale convolutional neural network (CNN) for geospatial object detection in high-resolution satellite images (from Google Earth). Bosch [21] and Ji *et al.* [35] proposed using 3-D CNNs to combine spatial and temporal information for motion recognition. Koga *et al.* [20] presented a CNN-based method for vehicle detection from aerial images combined with hard example mining in the training process. Al-Shakarji *et al.* [36] used DL combined with flux tensor spatiotemporal filtering to detect vehicles from aerial video. Li *et al.* [37] proposed a feature pyramid network combined with a saliency algorithm to predict oriented bounding boxes for objects from aerial images. However, for general or aerial images, the resolution is high enough to get vehicles' shape, color, or texture features to satisfy the demand of identification.

In satellite videos, the detailed information of tiny vehicles cannot be obtained for the limited resolution. An alternative method for detecting such small objects is to combine DL models with BGMs or feature fusion of multiframes. Zhang *et al.* [17] tried to use Fast R-CNN [38] to obtain coarse region proposals and a discrete histogram mixture model for vehicle detection in gray videos from SkySat-1. Ellenfeld *et al.* [39] introduced a CNN for multimodal motion segmentation, which fused the information from differencing of the current and its previous frames. Xiao *et al.* [40] proposed a two-stream detection network named dynamic and static fusion network (DSFNet) with the input of consecutive frames to detect moving object in satellite videos. Furthermore, research by LaLonde *et al.* [5] presented a deep-learning-based two-stage CNN approach to predict vehicles in aerial motion imagery, which utilized both spatial and temporal information of consecutive frames. However, it exploits Otsu thresholding to convert the corresponding output maps and needs to discriminate the motion state of the detected targets across several frames to remove the static ones subsequently.

Considering the above issues, we proposed a differential module in the stage of feature extraction. In this module, the operation of differencing is conducted on the high-level feature map instead of raw pixels, in which the former is purer in terms of irrelevant noises or local misaligned regions. Subsequently, it integrates the motion information from unlabeled adjacent frames immediately after we get the difference maps, which can facilitate the extraction of effective motion semantic features and avoid the accumulation of false peak responses. However, the interference and noise cannot be wholly suppressed. Furthermore, we introduced transformer to refine the extracted feature for the final classifier.

Transformer [23] has been widely used in natural language processing (NLP) [41] and computer vision (CV) [42], [43], in which self-attention is the critical component making it unique from the CNNs. Based on the self-attention mechanism, transformer can capture both local and global

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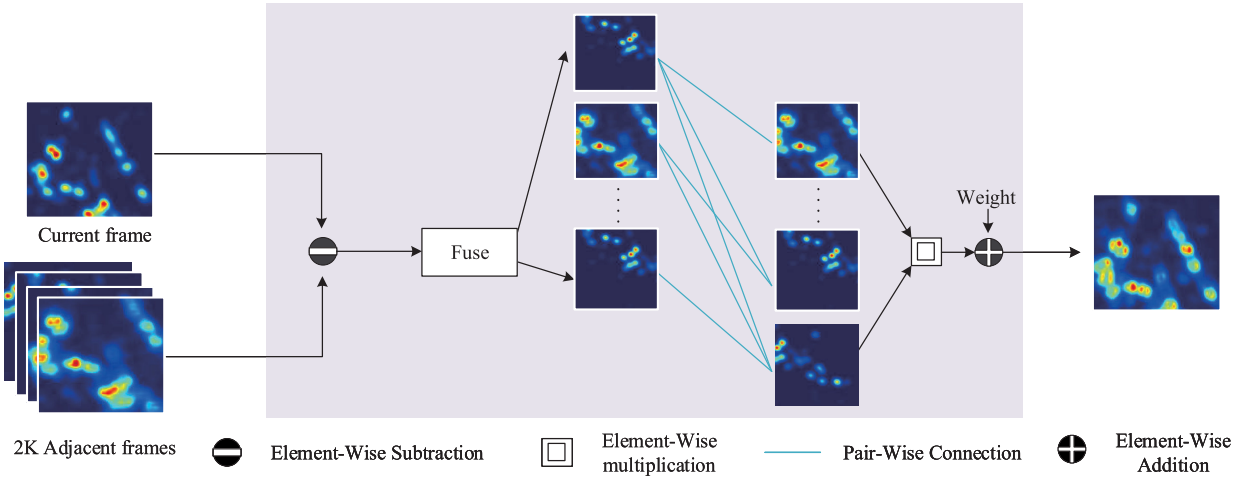


Fig. 3. Architectural details of differential module.

interactions. Some prior works [42], [44] demonstrated strong performance on image classification, object detection, or semantic segmentation by directly applying the standard transformer encoder [23] in NLP. However, it is intractable to adapt the encoder to high-resolution images or feature maps as the computations and memory cost is quadratic to image size. As presented in Section I, satellite videos exhibit a unique type of local misalignment and intensity variations [8]. Small and uneven motions on some stationary background objects are primary sources of false alarms in moving object detection. Although we use the differential module to weaken the interference of cluttered background, it is challenging for the classifier head of the proposed neural network model to distinguish the moving vehicles from false alarms. Thus, transformer is employed to get sophisticated features, and in addition, we reduce the dimension of key-value matrices considering the high computations. More details will be introduced in Section III.

III. Proposed Method

The proposed framework is based on the paradigm of object detection by key point estimation and scale prediction [45]–[47]. Referring to CenterNet [45], it uses Gaussian kernels to represent object center points. As shown in Fig. 2, the overall pipeline contains three main components: 1) the differential module to mitigate the background interference and enhance the semantic feature; 2) the transformer encoder to refine the feature, and 3) the head of classifier and bounding box regression.

Moving vehicles in the satellite video were captured with only several or tens of pixels spatially. Hence, the network's backbone (or feature extractor) should be lightweight with a low downsampling rate to allow for a relatively higher resolution output. In this article, the backbone is a Siamese network. The two streams of the network share the same parameters and architecture, which is a modified 18-layer residual network (ResNet18) [48]. Inspired by [45], we add three 4×4 transposed convolutional layers [49] with channels 256, 128, and 64 after ResNet18 to improve the resolution of output. Besides, a 3×3 convolutional layer with batch normalization and ReLU activation [50] is added before each transposed convolutional layer. The channel of the three convolutional layers is also 256, 128, and 64.

A. Differential Module

From Fig. 1, it can be observed that the moving vehicles appear as spots and the scenes of satellite video are complex. To alleviate the interference of noise and point-shaped analogs in the background, we propose a differential module to weaken the features of static area based on the motion information from adjacent $2K$ frames. Fig. 3 overviews the pipeline of this module.

The input of this module is a set of feature maps denoted by $F = \{f_{t-K}, f_{t-K+1}, \ldots, f_t, \ldots, f_{t+K}\}$, in which f_t indicates the feature map of the current frame and the others indicate that of the adjacent $2K$ frames. First, we calculate the pixel-wise difference between the feature maps

$$d_k = (f_t - f_k)^2, \quad k \in \{t-K, t-K+1, \ldots, t-1, t+1, \ldots, t+K\} \quad (1)$$

where d_k indicates the differencing map between the current frame and the kth neighboring frame. After we get the set of differencing maps $D = \{d_{t-K}, d_{t-K+1}, \ldots, d_{t-1}, d_{t+1}, \ldots, d_{t+K}\}$, a convolutional layer with kernel size 3×3 followed by ReLU activation is adopted to fuse each differencing map independently and parallelly. To a certain extent, most of the background objects are suppressed based on the differencing operation. However, when the frame is undergoing serious intensity variation and local misalignment, the differencing map of each pair frames is not that reliable. Therefore, the relationship between the generated maps (i.e., temporal information) should be considered to facilitate the feature refinement. In this way, the final map corresponding to the current frame receives weighted messages from other frames, which can be seen as temporal-contextual information. Let $\tilde{D} = \{\tilde{d}_{t-K}, \tilde{d}_{t-K+1}, \ldots, \tilde{d}_{t-1}, \tilde{d}_{t+1}, \ldots, \tilde{d}_{2K}\}$ be the differencing maps generated by previous fusing operation. We implement the pairwise message passing by weighted

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elementwise multiplication operations as follows:

$$f^* = \frac{1}{2} \sum_{i=t-K, i\neq t}^{t+K} \sum_{j=t-K, j\neq t}^{t+K} w(i,j)\langle \tilde{d}_i, \tilde{d}_j \rangle \tag{2}$$

where $w(i, j)$ is the element of weight matrix at (i, j) and $\langle \tilde{d}_i, \tilde{d}_j \rangle$ is the elementwise multiplication between $\tilde{d}_i$ and $\tilde{d}_j$. We regard w as a specific weight matrix, which represents the temporal correlation measurement, and it is a reasonable assumption that closer is stronger. Thus, we introduce sine function to calculate w, which is defined as

$$w(i,j) = \begin{cases} \sin\left(\frac{\pi}{2|i-t|}\right) \sin\left(\frac{\pi}{2|j-t|}\right), & i \neq j \\ 0, & i = j. \end{cases} \tag{3}$$

$w(i, i)$ is set as 0 to prevent message passing from $\tilde{d}_k$ to itself. The temporal-contextual information integration is achieved by message passing from adjacent frames, which can enhance the responses of regions that are truly moving vehicles and suppress the noisy background. It is worth noting that we choose the nearest $2K$ adjacent frames around the current one when it is in the first or last K frames, which do not have enough previous or next neighbors. Since all the operations of message passing are differentiable, the whole network can be trained in an end-to-end manner.

B. Transformer Encoder

Generally, the transformer encoder consists of N layers of multihead self-attention (MSA), and multilayer perceptron (MLP) blocks [23]. As presented in Fig. 4, we follow ViT [44] to adopt the normalization layer [51] in the residual unit before the MSA/MLP, which has been shown to be more competent than the counterpart [52]. The MLP block consists of two linear transformation layers with ReLU activation in between. The traditional form of MHA with M heads is a function defined as

$$\text{Head}_i = \text{Atten}(QT_1, KT_2, VT_3) \tag{4}$$

$$\text{MHA}(Q, K, V) = [\text{Head}_1; \text{Head}_2; \ldots; \text{Head}_M]L \tag{5}$$

where query Q, key K, and value V are the input of MHA and have a dimension of d. If $Q = K = V$, it is the MSA mechanism. $T_1, T_2, T_3 \in \mathbb{R}^{d\times d'}$ are the weight matrixes to compute the so-called query, key, and value embedding. d' is the dimension of each head and equal to (d/M). [;] denotes concatenation on the channel axis, and $L \in \mathbb{R}^{d\times d}$ is a linear projection matrix. The M heads are concatenated and projected by L to the final values. Atten$(\cdot)$ is the attention operation that is calculated as

$$\text{Atten}\left(\tilde{Q}, \tilde{K}, \tilde{V}\right) = \text{SoftMax}\left(\frac{\tilde{Q}\tilde{K}^T}{\sqrt{d'}}\right)\tilde{V} \tag{6}$$

where $\tilde{Q} = QT_1$, $\tilde{K} = KT_2$, and $\tilde{V} = VT_3$.

Since the small size of vehicles in satellite video, a high-resolution feature map is needed. Thus, before the transformer encoder, the stage of patch embedding comprises a 1×1 convolutional layer and a normalization layer. The output of patch embedding will be reshaped into a 1-D sequence of embedding tokens, and the sequence length is equal to the output feature map size. The dimension of embedding tokens is set to 32 to save computation and corresponds to MSA's input dimension d. Then, we add positional encodings that can be learned during the training process to the tokens. Positional encoding can encode the information about the relative or absolute position. Such position information may benefit context modeling. The transformer encoder receives the sequence of these tokens as input, and the output is reshaped back to the so-called refined feature map.

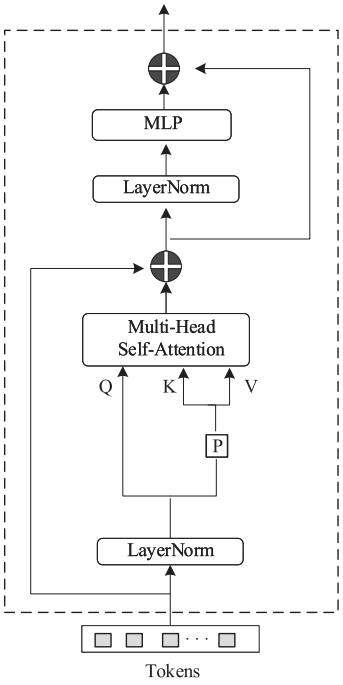


Fig. 4. Architecture of transformer encoder. P denotes max pooling.

Due to the computation of self-attention globally, transformer has quadratic computation complexity concerning input image size. However, we seek to get a high-resolution feature map for the tiny size of vehicles. Consequently, we leverage the max-pooling operation to reduce the spatial scale of K and V matrices before attention operation. The distribution of moving vehicles is sparse in satellite video, and we can select the regions of peak response by the process. In this way, the interaction could concentrate on the regions of actual moving vehicles instead of redundant information.

C. Head of Classifier and Bounding Box Regression

After the first two modules, we get refined feature map for both object classification and bounding boxes regression. There exist three separate heads to simultaneously predict the category of feature, center point offset, and object size wh (width and height). For an input image of width W and height H, a center point heatmap $P \in [0, 1]^{(W/R)\times(H/R)\times C}$ is produced in the classification process, where R is the output stride and C is the object categories. Following [45], the ground truth center points are splatted on a heatmap $Y \in [0, 1]^{(W/R)\times(H/R)\times C}$ using a Gaussian

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kernel $Y_{xyc} = \exp(-((x-\tilde{p}_x)^2 + (y-\tilde{p}_y)^2/2\sigma_p^2))$, where $(\tilde{p}_x, \tilde{p}_y) = \lfloor (p/R) \rfloor$ is the low-resolution equivalent point of the ground truth center point p of class c and σ_p is an object size-adaptive standard deviation [53]. A single-scale prediction map $S \in \mathbb{R}^{(W/R)\times(H/R)\times 2}$ is produced in regression process for all classes to limit the computational cost. For center point p_k, the scale prediction is $S_k = (w_k, h_k)$, where w_k and h_k correspond to the width and height of the object centered at p_k, respectively. $(x_1^k, y_1^k, x_2^k, y_2^k)$ is the final bounding box of object corresponding to p_k, and $S_k = (x_2^k - x_1^k, y_2^k - y_1^k)$. To recover the discretization error caused by the output stride, the local offset map $O \in \mathbb{R}^{(W/R)\times(H/R)\times 2}$ is also predicted. For center point p_k, the offset prediction is $O_k = (\delta_x^k, \delta_y^k)$. We can calculate the predicted bounding box coordinates, i.e.,

$$x_1^k = p_x^k + \delta_x^k - \frac{w_k}{2} \tag{7}$$

$$y_1^k = p_y^k + \delta_y^k - \frac{h_k}{2} \tag{8}$$

$$x_2^k = p_x^k + \delta_x^k + \frac{w_k}{2} \tag{9}$$

$$y_2^k = p_y^k + \delta_y^k + \frac{h_k}{2}. \tag{10}$$

At the inference phase, by applying a max-pooling operation with kernel size 3×3, the top 100 peaks whose value is greater or equal to its eight neighbors in the heatmap will be extracted. In general, the overall training objective is expressed as

$$L = L_{\text{cls}} + \lambda_1 L_{\text{wh}} + \lambda_2 L_{\text{offset}} \tag{11}$$

where L_{cls} indicates the classification loss, which is implemented by focal loss [54]. L_{wh} is the offset loss of wh, and we use Smooth-L1 loss [55] to train it to reduce the oscillation of the regression values near ground truth. L_{offset} indicates the offset loss of center points from feature misalignment, which is also implemented by Smooth-L1 loss. λ_1 and λ_2 are the tradeoff parameters. Those parameters signify the weights for corresponding loss. In this case, accurate location is of relatively more importance, and the size of vehicles changes slightly. Empirically, we set $\lambda_1 = 0.1$ and $\lambda_2 = 1$

$$L_{\text{cls}} = \frac{-1}{N} \sum_{xyc} \begin{cases} \left(1-\hat{Y}_{xyc}\right)^{\alpha} \log\left(\hat{Y}_{xyc}\right), & \text{if } Y_{xyc} = 1 \\ \left(1-Y_{xyc}\right)^{\beta} \left(\hat{Y}_{xyc}\right)^{\alpha} & \\ \times \log\left(1-\hat{Y}_{xyc}\right), & \text{otherwise} \end{cases} \tag{12}$$

where α and β are hyperparameters of the focal loss, and we set $\alpha = 2$ and $\beta = 4$ in our experiments following Law and Deng [53]:

$$L_{\text{wh}} = \frac{1}{N} \sum_{k=1}^{N} \text{Smooth}_{L1}\left(S_k - \hat{S}_k\right) \tag{13}$$

in which

$$\text{Smooth}_{L1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise} \end{cases} \tag{14}$$

where $\hat{S}_k$ is the scale prediction at center point p_k and S_k is the corresponding values of ground truth

$$L_{\text{offset}} = \frac{1}{N} \sum_{k=1}^{N} \text{Smooth}_{L1}\left(O_k - \hat{O}_k\right) \tag{15}$$

where $\hat{O}_k$ is the offset prediction of center point p_k and $O_k = (p/R) - \lfloor (p/R) \rfloor$ is the ground truth.

IV. Experimental Results and Analysis

Cameras on the satellite platform aim to capture a fixed region despite moving along the orbits. The satellite video has a relatively static background during a short period. However, there still exists much misalignment in local regions. In this section, we evaluate the detection performance of the proposed framework on satellite videos. The experiments are implemented in the Pytorch environment on a machine with NVIDIA Titan Xp GPUs.

A. Dataset

The experimental dataset is constructed from the satellite videos captured over San Diego by JiLin-1 on May 23, 2017, whose spatial resolution is around 1 m and is taken at 10 frames/s. The video sequences contain 311 frames, and the vehicles in these videos are overly small. For diversity, we select videos of various scenarios for training and evaluation in the experiments, including the overpass, crossroad, coastal road, sea bridge, and residential area. Compared to the complex background, moving vehicles occupy only a few pixels and have many distractors or similar surroundings. Moreover, it is challenging when moving vehicles undergo severe motion blur on the low-contrast roads. In addition, the boundary's displacement of tall buildings in the residential area or the fluctuation of seawater around the bridge may significantly raise the risk of detection.

The ground truth was generated manually by tagging all moving vehicles with boundary boxes. Training a neural network requires a large number of labeled samples. The moving vehicles are very tiny, and some are too dim to notice. Labeling work is arduous and tedious. However, our approach only needs annotations of several chosen frames (about 1/5) for every video. Since the neighboring frames do not need information for supervision, it could save us much time in practical application. As listed in Table I, we give detailed information about the dataset.

B. Experimental Setups

In the experiments, we use videos (No.1 to No.10) to train the models and the remaining videos for performance evaluation against existing state-of-the-art methods. Each input image is cropped to 224×224 for the purpose of data augmentation. The 224×224 crop is picked from the center or the four corners from the entire image. The data are also augmented by horizontal flipping and random scaling. The parameters of our backbone ResNet18 are initialized with ImageNet pretrain. Other convolutional layers and linear layers are randomly initialized using Pytorch's default settings. Considering the high computation consumption of transformer, we set the number of encoder layers to 2 and the heads to 4. The positional encodings are initialized to zero. Adam [56] is used to optimize the overall objective with an initial learning rate of $1.25e^{-3}$. The batch size is set to 36. It is trained for 160 epochs with the learning rate dropped at 70, 120, and 150 epochs by a factor of 0.1.

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TABLE I
DETAILED INFORMATION FOR THE DATASET

Video	Frame Size	#vehicles	Video	Frame Size	#vehicles	Video	Frame Size	#vehicles
No.1	300×480	34	No.6	410×340	64	No.11	350×330	38
No.2	400×410	22	No.7	380×300	27	No.12	440×390	44
No.3	490×480	18	No.8	360×290	7	No.13	490×440	27
No.4	420×410	25	No.9	610×560	38	No.14	390×480	25
No.5	490×480	65	No.10	450×370	19	No.15	460×400	41

Note: #vehicles is the average number of moving vehicles per frame.

TABLE II
DETECTION PERFORMANCE COMPARISON OF OUR PROPOSED MODEL WITH DIFFERENT PARAMETERS K

Model	No.11			No.12			No.13			No.14			No.15			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
K=1	76.66	63.03	69.18	67.31	65.04	66.16	53.20	66.47	59.10	58.38	67.24	62.50	58.88	93.32	72.20	62.89	71.02	65.83
K=2	82.71	66.91	73.97	71.90	61.10	66.06	79.61	52.10	62.99	58.56	62.50	60.47	61.58	94.25	74.49	70.87	67.37	67.60
K=3	83.69	77.16	80.29	69.66	75.07	72.26	80.89	82.57	81.72	59.92	76.33	67.13	63.62	95.33	76.31	71.56	**81.29**	**75.54**
K=4	85.83	70.36	77.32	73.18	60.57	66.28	80.01	75.38	77.63	62.85	70.49	66.45	66.46	93.76	77.78	**73.67**	74.11	73.09
K=5	85.50	66.54	74.84	71.93	67.15	69.46	81.34	68.64	74.45	59.53	71.97	65.16	65.53	89.19	75.55	72.77	72.70	71.89

C. Evaluation Metric

Quantitative results are evaluated on recall, precision, and F1 scores given by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

where true positive (TP) denotes the number of correct detections, false negative (FN) denotes the number of missed detections, and FP denotes the number of false alarms. A reference vehicle is considered TP if the intersection over union (iou) is greater than a threshold and FN otherwise. If multiple detections correspond to a reference vehicle, we choose the one with the largest iou. The detection does not intersect with any reference vehicle, which is regarded as FP. Due to the low spatial resolution, the majority of the vehicles in satellite remote sensing videos only occupy 2–20 pixels and appear as spots without any textures. In contrast with general objects, the conventional iou metric is quite sensitive to the predictions and cannot adapt to the tiny vehicles in satellite videos [3]. Tiny shifts of the predicted bounding box (i.e., 1 or 2 pixels) will cause a large fluctuation of the IoU score [57]. Therefore, the threshold of iou is set as 0.2, which could satisfy the requirements for locating moving targets in a wide area.

D. Performance Evaluations

1) Parameter K: At first, the output of the differential module was determined by the parameter K. It would influence the detection performance based on information integration from different adjacent frames. The proposed model with different K values was compared with each other. Table II shows the quantitative metric of different models. As described in Section III, we exploit the operation of differencing between feature maps of the current frame and its adjacent frame to alleviate the interference of background areas. Because of the similarity of background, we can get a sparse feature map, including much motion information. However, the obtained feature map of each pair of frames always contains the responses of noisy background and stationary vehicles, which is mainly caused by the extreme intensity variation and local misalignment of the complex scenes. Thus, we integrate K feature maps to facilitate the extraction of enhanced semantic features and suppress the responses of mistaken regions. Nevertheless, at the same time, it might introduce extra interference when the temporal context relevance is not that strong. As shown in Fig. 5, it displays the integrated feature maps across a different parameter K. It can be found that our module learns to highlight areas of the moving vehicles gradually and activate more regions as K increases. Consequently, the recall score, as well as precision, might be improved due to the assistance from neighboring frames. It is consistent with the quantitative result in Table II. For satellite video, on the other hand, the frame will undergo more seriously local variation with a longer interval of time. Therefore, the extra noise might be introduced, and we can find many more false active regions on the feature maps when $K = 4$ or 5. Correspondingly, due to the presence of distractors, it will degrade the performance to some extent. To make an intuitive explanation, we provide the trend of the three criteria with K in Fig. 6. It shows that the relationship between model performance and the number of integrated frames is not positively correlated. The recall score grows nearly to saturation as we combine enough short-term consecutive frames with the current frame. The best result is obtained when increasing K to 3, demonstrating that the temporal contextual information can be well-integrated by three pairs of neighbor frames.

2) Weighted Fusion: Based on the operation of differencing, we can get the efficient motion feature of candidate targets

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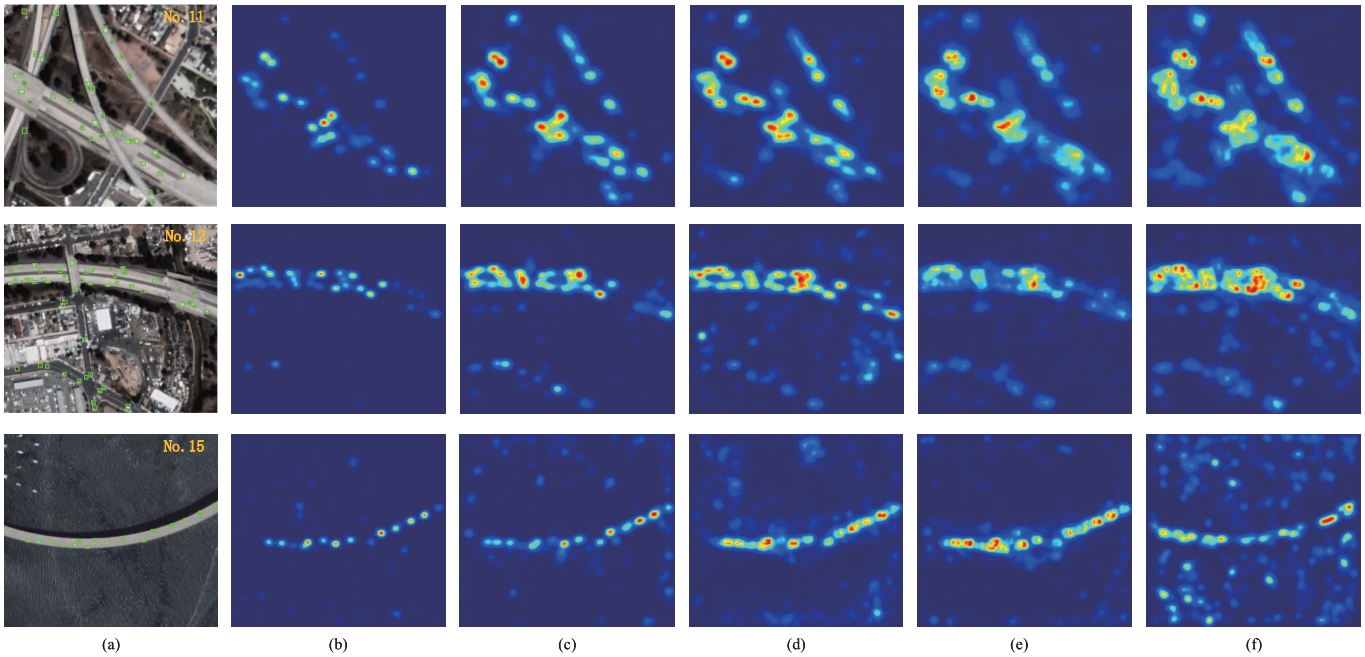


Fig. 5. Examples of the integrated feature maps with different parameters K. (a) Examples of the complex scenes in JiLin-1 satellite images. (b) $K = 1$. (c) $K = 2$. (d) $K = 3$. (e) $K = 4$. (f) $K = 5$.

TABLE III

DETECTION PERFORMANCE COMPARISON OF OUR PROPOSED MODEL WITH AND WITHOUT THE WEIGHTED FUSION

Method	No.11			No.12			No.13		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
With	83.69	77.16	80.29	69.66	75.07	72.26	80.89	82.57	81.72
Wout	72.11	81.51	76.52	62.08	69.52	65.59	70.00	55.24	61.75
Method	No.14			No.15			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
With	59.92	76.33	67.13	63.62	95.33	76.31	71.56	81.29	75.54
Wout	57.54	50.09	53.56	66.48	82.02	73.44	65.64	67.68	66.17

Note: the numbers marked in red indicate a decrease.

TABLE IV

DETECTION PERFORMANCE COMPARISON OF THE WEIGHTED FUSION PART BEFORE AND AFTER THE BACKBONE NETWORK

Method	No.11			No.12			No.13		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
AFT	83.69	77.16	80.29	69.66	75.07	72.26	80.89	82.57	81.72
BEF	81.65	76.44	78.96	75.11	61.24	67.47	83.73	61.74	71.07
Method	No.14			No.15			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
AFT	59.92	76.33	67.13	63.62	95.33	76.31	71.56	81.29	75.54
BEF	70.26	57.32	63.13	57.68	62.26	59.88	73.69	63.80	68.10

Note: the numbers marked in red indicate a decrease.

in the current frame. Compared to existing methods for moving object detection, a weighted fusion part is added to integrate the motion feature on high-level maps in the proposed differential module. Therefore, we add two comparative studies to validate the effectiveness of this process. In the first experiment, we just keep the differencing operation between the current frame and its adjacent frame to extract motion information, and the rest of the network structure remains the same. Based on this ablation study, we can validate the effectiveness of the weighted fusion part. Table III shows the quantitative result of the test videos. It validates the effectiveness of the weighted fusion part for superior performance in any criterion. Then, in the second experiment, we conduct the differencing operation and the weighted fusion between raw input frames instead of high-level feature maps. Table IV presents the quantitative result. During the first stage of semantic feature extraction by the backbone network, it will suppress some regions of distractors or filter out unrelated noise. Hence, when the motion features of adjacent frames are integrated, there will be no accumulation of peak responses in irrelevant regions. Consequently, it will contribute to better performance in precision with little damage to the recall score.

3) Transformer Encoder: In addition, we perform an ablation study on transformer encoder to validate its effectiveness in context modeling. As reported in Table V, we can observe significant drops in precision on the test dataset when removing the transformer encoder. Specifically, it mainly decays the model's performance to distinguish objects from interference or distractors and shows no noticeable impact on the recall score. It indicates that the transformer encoder can well refine the features of moving objects for the classifier by local and global interactions. As shown in Fig. 7, we give some visual results of the feature maps with the refinement of transformer encoder for comparison. Based on the integration of feature maps from adjacent frames, more regions on the map corresponding to the moving vehicles were highlighted, which might facilitate the extraction of enhanced semantic features.

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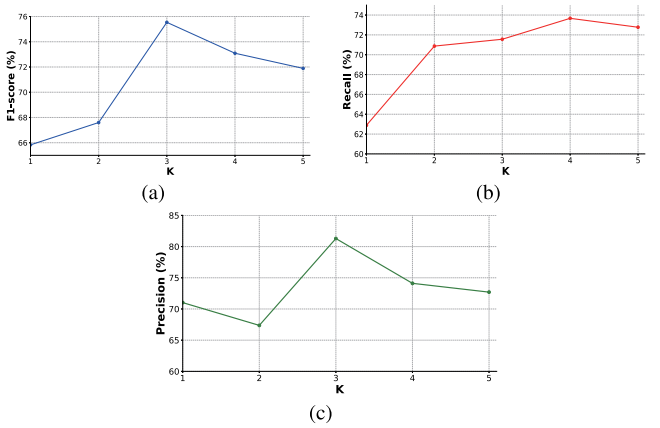


Fig. 6. Average performance of our proposed approach with a different number of integrated frames on the test dataset. (a) F1. (b) Recall. (c) Precision. Since the neighbor frame in a long interval of time is relatively unreliable, it might introduce more interference to degrade the precision and F1-score if further increasing the parameter K.

TABLE V
DETECTION PERFORMANCE COMPARISON OF OUR PROPOSED MODEL WITH AND WITHOUT TRANSFORMER ENCODER

Method	No.11			No.12			No.13		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
With	83.69	77.16	80.29	69.66	75.07	72.26	80.89	82.57	81.72
Wout	83.77	56.33	67.36	69.35	53.57	60.45	78.55	55.50	65.04
Method	No.14			No.15			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
With	59.92	76.33	67.13	63.62	95.33	76.31	71.56	81.29	75.54
Wout	60.94	51.68	55.93	65.16	91.30	76.05	71.55	61.68	64.97

Note: the numbers marked in red indicate a decrease.

However, it can be found that the activated areas seem to stick together, especially when dense vehicles are moving on the roads. Because of the very small size of vehicles, the dense scenes make it difficult to distinguish the regions of peak response on the extracted feature maps. Consequently, it will hinder the location of targets. On the contrary, the active regions of the feature on the maps are separated well and have sharper peak responses after transformer encoder. We can get the refined semantic feature for object classification and bounding boxes regression.

In summary, an effective feature map can be obtained by integrating a reasonable number of adjacent frames. Then, based on the transformer encoder to refine the semantic feature, we will get the best performance. In other words, it demonstrates that the differential module and transformer encoder are complementary for the effectiveness of our proposed framework. The combined effect of these two modules improves the detection performance of the model for moving vehicles.

E. Comparisons

To verify the superior performance of our proposed approach, we conduct comparative experiments on tiny moving object detection with several state-of-the-art methods, which are ViBe [11], RPCA [27], Saur *et al.* [15], Ellenfeld *et al.* [39], Zhang and Xiang [18], MOSNET [21], ClusterNet [5], and DSFNet [40]. Table VI shows the quantitative evaluation results. Postprocessing was applied to all segmentation-based approaches for removing extremely small or large blobs. Their performance is compared in terms of recall, precision, and F1 scores. According to Table VI, it can be observed that our approach outperforms others with a significant margin. Specifically, the proposed method has a clear advantage on recall across all the test videos. It is due to the combined effect of the differential module and transformer encoder, in which the former well suppresses the interference of background motion and enhances the motion semantic features of real low-resolution moving targets, while the latter refines the features for the classifier to distinguish nonvehicle moving targets effectively.

To observe the detecting details of these methods, we give some visual results of the test dataset in Fig. 8. Intuitively, our method performs much better even if the targets undergo

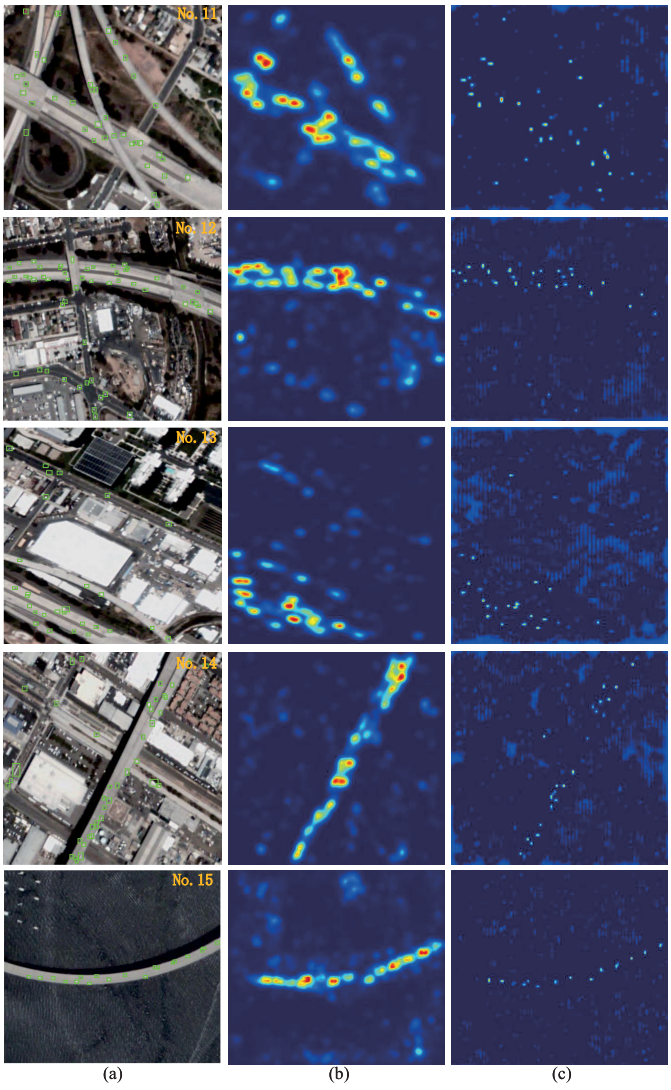


Fig. 7. Visual comparison of feature maps with the refinement of transformer encoder. (a) Examples of the complex scenes in JiLin-1 satellite images. (b) The corresponding integrated feature maps by the differential module with $K = 3$. (c) Refined feature maps after transformer encoder.

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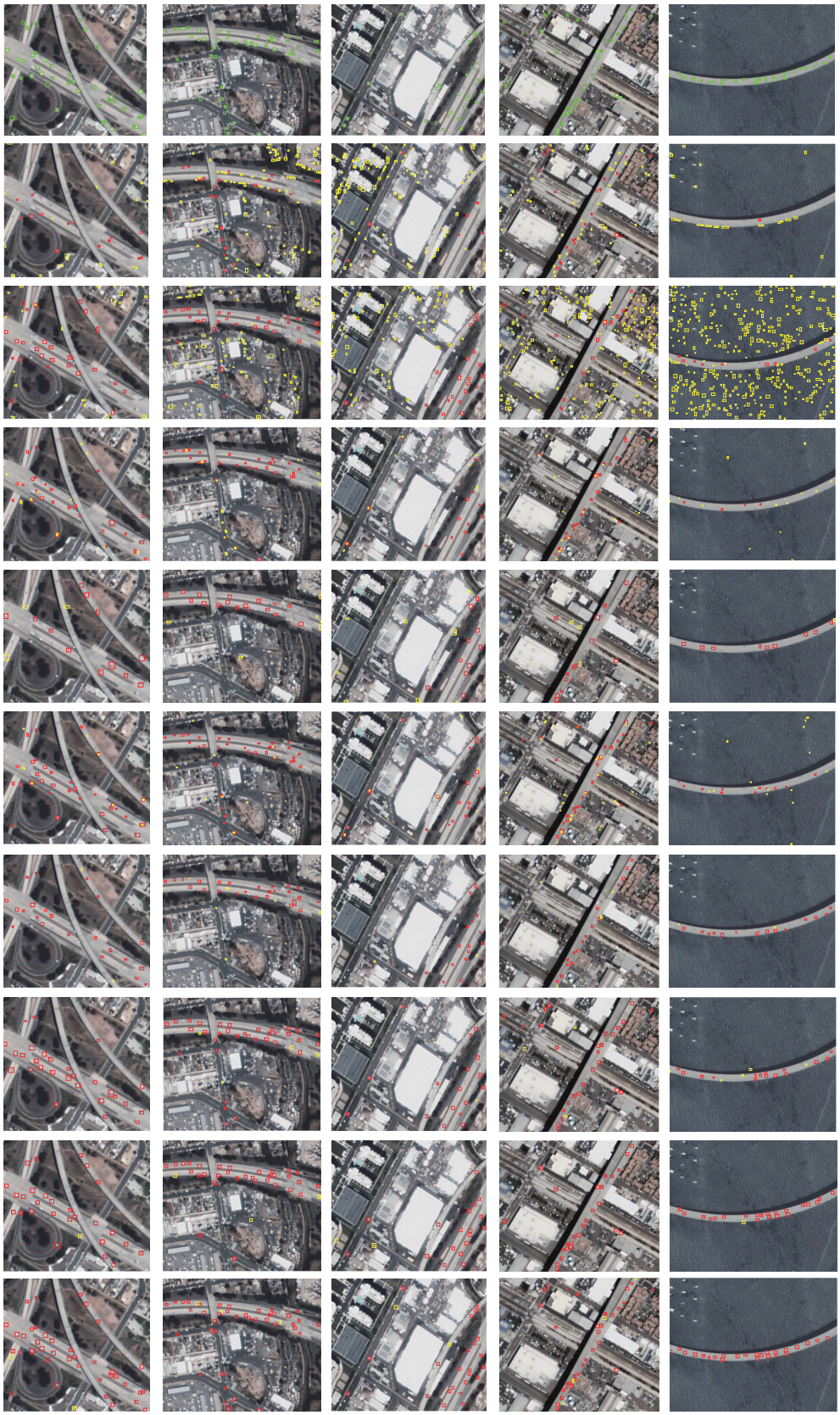


Fig. 8. Visual comparison of detection results on the test dataset. The top row is the ground truth of the example frames, and the green boxes indicate the moving vehicles. The result of ViBe [11] in row 2, RPCA [27] in row 3, Saur *et al.* [15] in row 4, Ellenfeld *et al.* [39] in row 5, Zhang and Xiang [18] in row 6, MOSNET [21] in row 7, ClusterNet [5] in row 8, DSFNet [40] in row 9, and ours in row 10. The red boxes indicate the positive targets, and false alarms are shown in yellow boxes.

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TABLE VI
DETECTION PERFORMANCE COMPARISON WITH THE STATE-OF-THE-ART ALGORITHMS

Method	No.11			No.12			No.13		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
ViBe [11]	33.03	22.43	26.72	38.77	18.26	24.82	30.63	3.76	6.70
RPCA [27]	70.39	52.13	59.90	71.48	32.86	45.02	66.79	16.07	25.91
G. Saur [15]	67.45	63.01	65.15	57.65	48.64	52.76	59.86	45.35	51.60
Ellenfeld [39]	36.41	65.76	46.87	38.38	76.44	51.10	42.51	57.97	49.05
Zhang [18]	73.79	58.95	65.54	66.69	40.57	50.45	71.36	40.09	51.34
MOSNET [21]	64.95	85.59	73.85	53.31	80.75	64.23	48.29	72.51	57.97
ClusterNet [5]	67.47	88.33	76.50	65.72	79.27	71.86	69.88	82.47	70.89
DSFNet [40]	74.27	79.24	76.67	65.56	78.88	71.61	69.79	75.40	72.49
Ours	83.69	77.16	80.29	69.66	75.07	72.26	80.89	82.57	81.72
Method	No.14			No.15			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
ViBe [11]	37.54	10.66	16.60	1.81	1.19	1.44	28.36	11.26	15.26
RPCA [27]	56.95	20.96	30.64	52.51	3.24	6.10	63.62	25.05	33.51
G. Saur [15]	45.38	46.08	45.72	31.92	43.68	36.89	52.45	49.35	50.42
Ellenfeld [39]	26.27	66.67	37.69	32.58	87.65	47.50	35.23	70.90	46.44
Zhang [18]	52.88	37.12	43.62	43.78	35.91	39.45	61.70	42.53	50.08
MOSNET [21]	43.92	83.80	57.64	51.53	83.15	63.63	52.40	81.16	63.46
ClusterNet [5]	52.10	84.54	64.47	49.53	87.75	63.32	60.94	**84.47**	69.41
DSFNet [40]	53.13	81.46	64.32	62.07	89.43	73.28	64.96	80.88	71.67
Ours	59.92	76.33	67.13	63.62	95.33	76.31	**71.56**	81.29	**75.54**

severe motion blur on the low-contrast background area. On the contrary, it shows that MOSNET [21], ClusterNet [5], and DSFNet [40] well reduced the FP detections in the cluster background and presented a good performance in precision, but it failed to recall many unclear moving vehicles from the low-contrast roads. Zhang and Xiang [18] could identify more moving targets but introduced a large number of false motions. Ellenfeld *et al.* [39] added motion information from the raw input frames to the high-level network feature maps, which would make the model pay more attention to motion regions. It obtained a good precision score but lost a lot of targets with poor features. In addition, traditional methods for moving object detection, such as ViBe [11] and RPCA [27], failed to distinguish the unrelated motion of background objects, including the false objects on the top of buildings and the boundary of roads and bridges. ViBe [11] missed most of the targets in a scene of the sea bridge, which suffered a very significant displacement. Because of the fluctuation of seawater, there have been a lot of false alarms proposed by RPCA [27]. In particular, Saur *et al.* [15] introduced a modified image differencing method to suppress unrelated motion by using the minimum differences of each pixel in small neighborhoods, which shows an appreciable performance in precision.

F. Evaluation on Public Satellite Video Datasets

In this section, the proposed framework was also evaluated on two public satellite video datasets. The first dataset was captured by Jilin-1 satellite platforms and well annotated by Xiao *et al.* [40] and Yin *et al.* [57]. It is called VISO and contains a great number of moving objects (1 646 038) with different scales. For the challenge of detecting very small objects with low spatial resolution, we follow Xiao *et al.* [40] to concentrate on the detection task of the car (generally less than 50 pixels), which occupies nearly 90% of the VISO dataset. The second dataset was provided by Ao *et al.* [3] and

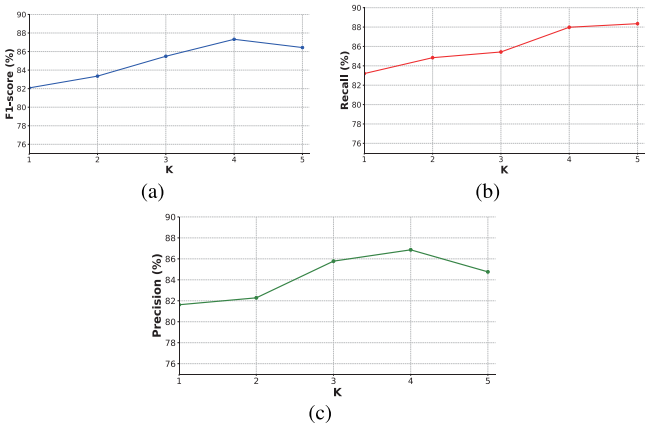


Fig. 9. Average performance of our proposed approach with a different number of integrated frames on the VISO dataset. (a) F1. (b) Recall. (c) Precision.

captured by Jilin-1 satellite over Valencia. It consists of three video sequences and contains nearly 33 640 moving vehicles with very small scales. For brevity, we call it Valencia in this article.

1) VISO: In consistent with [40], we train our model with the 72 videos of 512 × 512 and evaluate metrics on the seven videos of 1024 × 1024. The data are augmented by random crop, horizontal flipping, and random scaling. The batch size is set to 48 and it is trained for 140 epochs with the initial learning rate of $1.25e^{-4}$, which dropped by a factor of 0.1 when the objective loss stops falling. All other hyperparameters are the same as in the previous experiment. Table VII shows the quantitative results of the test videos on recall, precision, and F1 scores with the different parameters K. In addition, we also provide the trend of the three criteria with K in Fig. 9 for an intuitive explanation. As mentioned above, based on information integration from different adjacent frames in the differential module, our method learns to highlight areas

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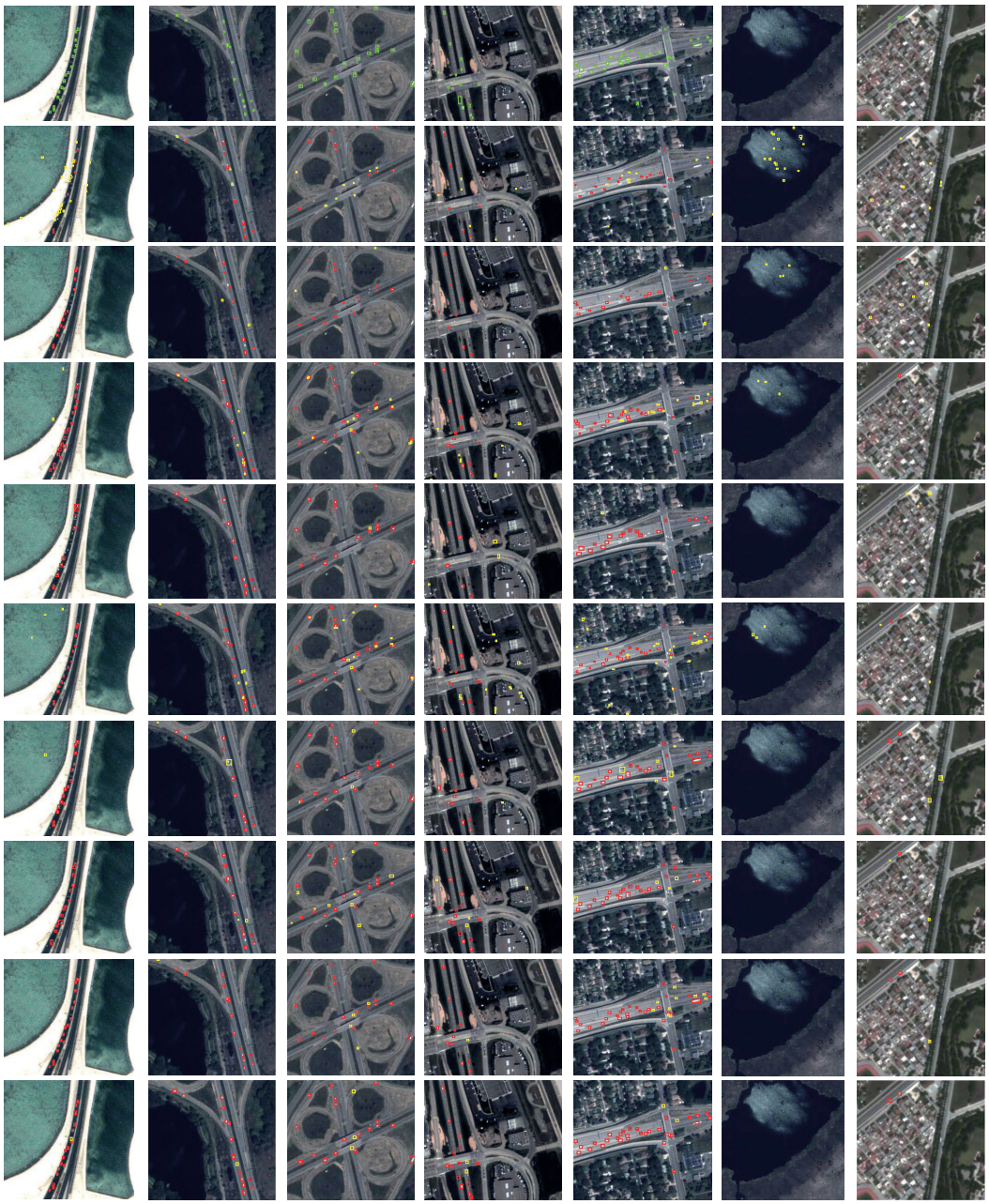


Fig. 10. Visual comparison of detection results on the VISO dataset. The top row is the ground truth of the example scenes, and the green boxes indicate the moving vehicles. The result of ViBe [11] in row 2, RPCA [27] in row 3, Saur *et al.* [15] in row 4, Ellenfeld *et al.* [39] in row 5, Zhang and Xiang [18] in row 6, MOSNET [21] in row 7, ClusterNet [5] in row 8, DSFNet [40] in row 9, and ours in row 10. The red boxes indicate the positive targets, and false alarms are shown in yellow boxes.

of the moving vehicles gradually and activate more regions as K increases. Consequently, the recall score, as well as precision, might be improved due to the assistance of adjacent frames. However, the extra noise might be introduced at the same time. It would degrade the performance of precision to some extent, and the recall score grows nearly to saturation. In this experiment, the best result is obtained when K increases to 4.

Moreover, we conduct comparative experiments on this dataset with state-of-the-art methods to verify the superior performance of our proposed approach. Table VIII shows the quantitative results, and we also give the visual comparison of detection results on example frames in Fig. 10. It can be observed that our approach presented the best performance. Specifically, MOSNET [21] and ClusterNet [5] have a good performance on precision but missed some blurry targets. DSFNet [40] obtained a relatively high recall score and suboptimal performance across all the methods. Ellenfeld *et al.* [39] failed to recall many weak targets. Zhang and Xiang [18] utilized the GMM to find out the moving regions and then employed the CNN model to classify the candidates, by which it could recall most of the low-contrast targets but performed

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TABLE VII

DETECTION PERFORMANCE COMPARISON ON THE VISO DATASET WITH DIFFERENT PARAMETERS K

Model	No.1			No.2			No.3			No.4		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
K=1	82.53	90.05	86.13	87.68	82.48	85.00	90.57	73.66	81.25	86.69	78.97	82.65
K=2	83.13	91.49	87.11	87.09	85.38	86.22	91.86	77.78	84.24	87.51	79.75	83.45
K=3	84.02	94.83	89.10	88.15	85.98	87.05	91.64	83.90	87.60	87.55	84.12	85.80
K=4	87.87	90.33	89.08	90.68	86.28	88.42	93.83	85.57	89.51	89.78	86.08	87.89
K=5	86.66	92.13	89.31	89.87	85.11	87.43	93.26	86.44	89.72	90.64	86.41	88.47
Model	No.5			No.6			No.7			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
K=1	84.40	79.46	81.86	77.50	76.04	76.76	73.06	90.67	80.92	83.20	81.62	82.08
K=2	85.73	80.16	82.85	76.20	85.33	80.51	82.33	76.05	79.07	84.84	82.28	83.35
K=3	85.18	79.31	82.14	79.01	86.00	82.35	82.49	86.31	84.36	85.43	85.78	85.49
K=4	87.78	80.16	83.80	82.80	87.13	84.91	83.77	92.49	87.92	88.07	**86.86**	**87.36**
K=5	88.54	76.61	82.15	81.73	84.47	83.08	87.72	82.12	84.83	**88.35**	84.76	86.43

TABLE VIII

DETECTION PERFORMANCE COMPARISON ON THE VISO DATASET WITH THE STATE-OF-THE-ART ALGORITHMS

Method	No.1			No.2			No.3			No.4		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
ViBe [11]	27.31	17.14	21.06	35.72	24.89	29.34	33.92	23.42	27.71	48.18	25.47	33.33
RPCA [27]	68.66	60.36	64.24	57.02	79.83	66.52	47.81	75.84	58.65	54.59	66.59	60.00
G. Saur [15]	76.61	42.85	54.96	73.01	56.12	63.46	78.73	52.79	63.20	80.75	60.62	69.25
Ellenfeld [39]	40.92	55.07	46.95	61.22	71.77	66.08	60.56	59.72	60.14	43.70	51.54	47.30
Zhang [18]	69.70	46.91	56.08	76.47	50.62	60.92	73.16	44.68	55.48	68.83	43.03	52.96
MOSNET [21]	77.01	83.71	80.22	77.83	86.89	82.11	84.12	86.93	85.50	78.07	85.95	81.82
ClusterNet [5]	70.47	73.51	71.96	84.21	75.34	79.53	89.22	65.33	75.43	82.71	80.79	81.74
DSFNet [40]	91.94	91.73	91.84	87.18	83.70	85.40	93.75	79.43	86.00	82.85	90.62	86.56
Ours	87.87	90.33	89.08	90.68	86.28	88.42	93.83	85.57	89.51	89.78	86.08	87.89
Method	No.5			No.6			No.7			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
ViBe [11]	36.48	20.06	25.89	22.57	19.90	21.15	14.43	17.06	15.64	31.23	21.13	24.87
RPCA [27]	43.36	73.12	54.44	38.76	56.80	46.08	35.58	38.43	36.95	49.40	64.42	55.27
G. Saur [15]	72.73	58.92	65.10	76.96	51.27	61.54	35.24	46.66	40.15	70.58	52.75	59.67
Ellenfeld [39]	47.12	57.86	51.94	49.93	59.75	54.40	39.73	43.89	41.71	49.03	57.09	52.65
Zhang [18]	68.41	48.28	56.61	61.08	40.07	48.40	51.00	47.47	49.17	66.95	45.87	54.23
MOSNET [21]	74.40	81.73	77.89	68.95	82.82	75.25	77.40	71.86	74.53	76.83	82.84	79.62
ClusterNet [5]	76.50	73.55	74.99	76.31	65.44	70.45	69.54	72.95	71.20	78.42	72.42	75.04
DSFNet [40]	94.89	66.46	78.17	81.46	84.56	82.98	83.95	86.47	85.19	88.00	83.28	85.16
Ours	87.78	80.16	83.80	82.80	87.13	84.91	83.77	92.49	87.92	**88.07**	**86.86**	**87.36**

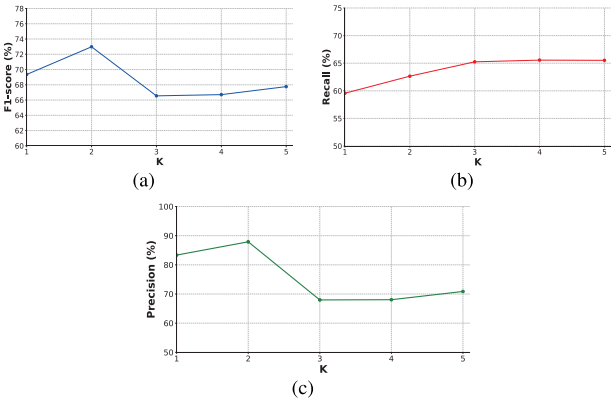


Fig. 11. Average performance of our proposed approach with a different number of integrated frames on the Valencia dataset. (a) F1. (b) Recall. (c) Precision.

not well in precision. Correspondingly, we can find some FP detections on fluctuating water in Fig. 10.

TABLE IX

DETECTION PERFORMANCE COMPARISON ON THE VALENCIA DATASET WITH DIFFERENT PARAMETERS K

Model	No.2			No.3			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
K=1	64.75	83.62	72.98	54.37	83.07	65.72	59.56	83.35	69.35
K=2	68.27	87.08	76.54	57.02	88.72	69.42	62.65	**87.90**	**72.98**
K=3	72.08	72.12	72.10	58.41	63.81	60.99	65.25	67.97	66.55
K=4	72.11	70.53	71.31	59.01	65.54	62.10	**65.56**	68.04	66.71
K=5	73.54	69.80	71.62	57.49	71.94	63.90	65.52	70.87	67.76

In conventional methods for motion detection, Saur *et al.* [15] got a higher recall compared to ViBe [11] and RPCA [27], but these algorithms are still very sensitive to the variation of the pixel and not that effective for objects with poor features, which are very common in satellite videos. Also, it can be observed that many distractors in the background area were falsely detected by these algorithms

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TABLE X
DETECTION PERFORMANCE COMPARISON ON THE VALENCIA DATASET WITH THE STATE-OF-THE-ART ALGORITHMS

Method	No.2			No.3			Average		
	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)	Rcll(%)	Prec(%)	F1(%)
ViBe [11]	43.25	3.97	7.27	44.37	1.95	3.74	43.81	2.96	5.51
RPCA [27]	33.19	14.18	19.87	55.24	9.49	16.19	44.22	11.84	18.03
G. Saur [15]	23.93	52.56	32.89	36.29	39.33	37.75	30.11	45.95	35.32
Ellenfeld [39]	35.22	68.75	46.58	45.24	72.34	55.67	40.23	70.55	51.13
Zhang [18]	53.99	62.89	58.10	58.40	51.21	54.57	56.20	57.05	56.34
MOSNET [21]	35.29	66.20	46.02	39.04	81.03	52.69	37.17	73.62	49.36
ClusterNet [5]	47.44	82.34	60.20	54.35	92.32	68.42	50.90	87.33	64.31
DSFNet [40]	61.80	91.03	73.62	52.36	77.14	62.38	57.08	84.09	68.00
Ours	68.27	87.08	76.54	57.02	88.72	69.42	**62.65**	**87.90**	**72.98**

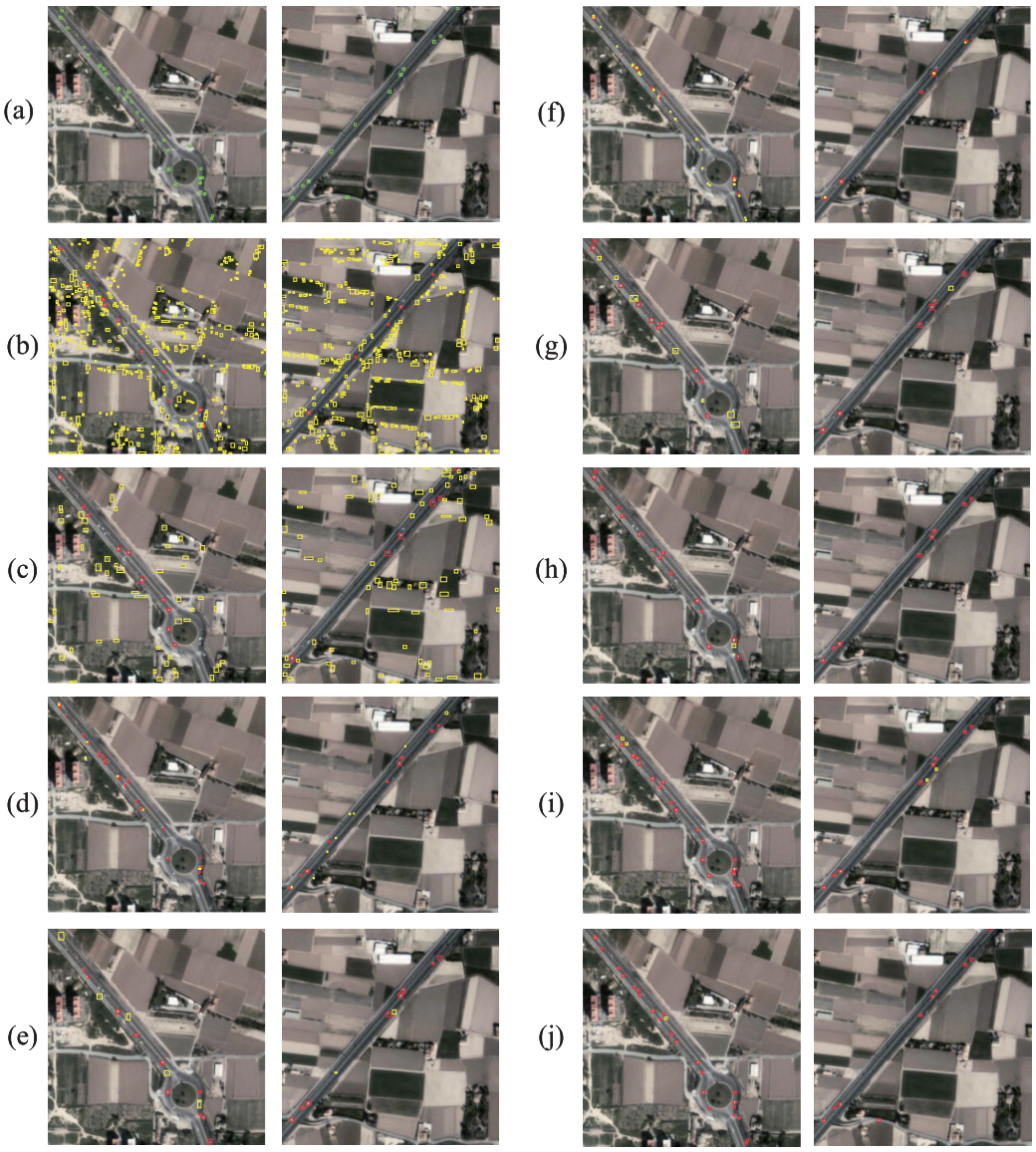


Fig. 12. Visual comparison of detection results on the Valencia dataset. (a) Ground truth of the example frames indicated by the green boxes. (b) ViBe [11], (c) RPCA [27], (d) Saur *et al.* [15], (e) Ellenfeld *et al.* [39], (f) Zhang and Xiang [18], (g) MOSNET [21], (h) ClusterNet [5], (i) DSFNet [40], and (j) ours. The red boxes indicate the positive targets, and false alarms are shown in yellow boxes.

from the visual results. On the contrary, our method can well suppress the interference of unrelated motion and extract sufficient motion semantic features.

2) Valencia: In the experiment, we choose one of the three video sequences to train the models and the remaining two for performance evaluation against existing state-of-the-art

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methods. For data augmentation, each input image is randomly cropped from the center or the four corners of the entire image. The data are also augmented by horizontal flipping and random scaling. The parameters are initialized with the same settings in previous experiments. The batch size is set to 32. It is trained for 120 epochs with the learning rate dropped by a factor of 0.1 when the objective loss stops falling.

Table IX presents the quantitative results on the test videos in terms of recall, precision, and F1 scores with the different parameters K. As shown in Fig. 11, it illustrated the similar trend of the three criteria with K and we got the best result when K increases to 2. It is consistent with the variation of quantitative results in previous experiments.

We further compare our approach with state-of-the-art methods on this dataset, which is reported in Table X. It can be found that our approach outperforms others on any criterion, especially the recall score. ClusterNet [5] got the suboptimal performance on precision, but it failed to distinguish some low-contrast targets, as shown in Fig. 12. Ellenfeld *et al.* [39] and MOSNET [21] could well avoid interference from noise and nontarget motion. However, they missed many blurry positive objects. In contrast, Zhang and Xiang [18] and DSFNet [40] obtained a good performance on recall. For conventional methods, Saur *et al.* [15] got relatively high precision compared to ViBe [11] and RPCA [27]. We can see lot of false alarms on the farmland boundaries from the visual results by ViBe [11] and RPCA [27].

V. Conclusion

To increase the accuracy of low-resolution moving vehicle detection in satellite video, we proposed an end-to-end framework that applies the differential module to integrate the temporal contextual information and refine the semantic features of moving objects by transformer. The proposed method contends with tiny targets of low quality in cluttered background. The experiments indicate that the differential module and transformer encoder can be well combined to improve the detection performance of moving vehicles. Despite the low contrast and resolution of satellite video, our framework shows superior performance in the complex scenes compared to many state-of-the-art algorithms. There is still some room for improvement, and many challenges remain. Due to the high computation consumption of transformer, we did not choose to increase the depth of encoder layers. We will investigate more about how to enhance the context modeling of transformer encoder in future work.

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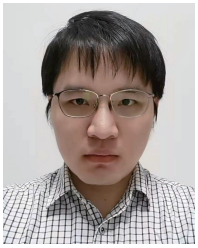
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